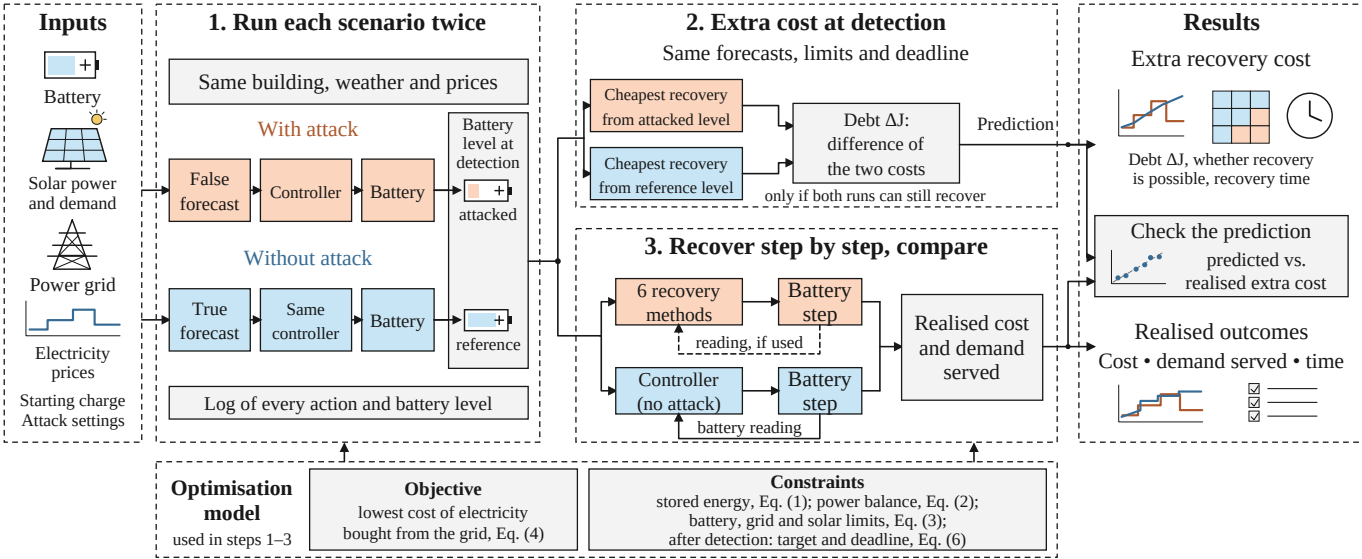
Inputs
Battery
Solar power and demand
Power grid
Electricity prices
Starting charge
Attack settings
1. Run each scenario twice
Same building, weather and prices
With attack
False forecast
Controller
Battery
Battery level at detection
attacked
Without attack
True forecast
Same controller
Battery
reference
Log of every action and battery level
2. Extra cost at detection
Same forecasts, limits and deadline
Cheapest recovery from attacked level
Cheapest recovery from reference level
Debt ΔJ: difference of the two costs
only if both runs can still recover
Prediction
3. Recover step by step, compare
6 recovery methods
Battery step
reading, if used
Controller (no attack)
Battery step
battery reading
Realised cost and demand served
Results
Extra recovery cost
Debt ΔJ, whether recovery is possible, recovery time
Check the prediction
predicted vs. realised extra cost
Realised outcomes
Cost • demand served • time
Optimisation model
used in steps 1–3
Objective
lowest cost of electricity bought from the grid, Eq. (4)
Constraints
stored energy, Eq. (1); power balance, Eq. (2);
battery, grid and solar limits, Eq. (3);
after detection: target and deadline, Eq. (6)